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Spatiotemporal evolution of marine new-quality productivity in China: Based on the coordinated regional development perspective

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Abstract

Marine new-quality productivity is the key driver of China's high-quality marine economic growth. Based on the three-dimensional reconfiguration of 'knowledgeable laborers, intelligent labor materials, and high-valued labor objects', this study defines its core connotation. We selected 11 coastal province-level regions in China from 2010 to 2023 to assess the development level of marine new-quality productivity and examined its spatiotemporal evolution from an economic geography perspective. The main findings are as follows: First, during the study period, the marine new-quality productivity in China's coastal areas remained in nascent stages of development but with a steady upward trend. The development levels of all components also maintained positive growth. Furthermore, the spatial pattern of 'eastern marine economic circle > northern marine economic circle > southern marine economic circle' remains largely stable, while regional imbalances are still prominent. Finally, the development of marine new-quality productivity is a slow process with a strong and stable level distribution, which tends to transfer upward. The Chinese case study offers valuable insights into the development of marine economies in other regions worldwide.

Keywords Chinese coastal areas, Marine new-quality productivity, Spatiotemporal evolution, Regional coordinated development

1 Introduction

The twenty-first century is often termed the century of oceans. Covering 71% of the Earth's surface, the ocean holds abundant biological, energy, and mineral resources. As constraints on land resources tighten and global climate change intensifies, the ocean has gradually become the core carrier for expanding humanity's development space. In recent years, China has repeatedly emphasized the 'Developing Marine Economy' strategy in important meetings and documents on policy measures that continuously promote high-quality development of the marine economy (Ding et al., 2019). In 2024, China's marine economy surpassed 10 trillion CNY for the first time (Fig. 1), gradually becoming an important driver of national economic development.

However, marine economy development still faces challenges such as resource-based transformation, regional imbalance, and inadequate green technological innovation (Wei et al., 2025). To accelerate maritime power construction, it is impossible to overcome existing technical problems or explore new marine spaces and resources through traditional and conventional productivity. Therefore, developing new-quality marine productivity is essential for promoting high-quality marine economic development and strengthening maritime power (Ji et al., 2024; Wang et al., 2025). Currently, the development of marine technology in China is fostering

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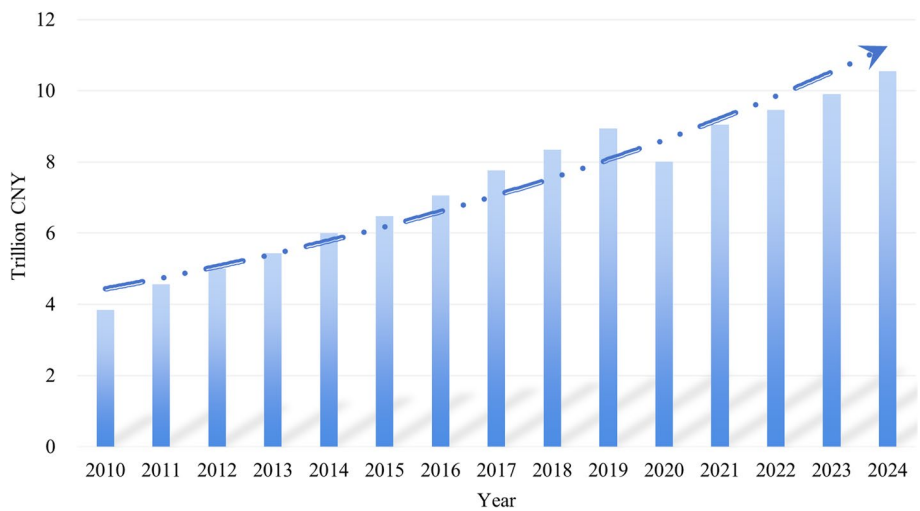

Fig. 1 China's Gross Ocean Product (GOP) from 2010 to 2024. Data source: Ministry of Natural Resources of P. R. China, China Marine Economy Statistical Bulletin (2010–2024), https://www.mnr.gov.cn/sj/sjfw/hy/gbgg/zghyjjtjgb/index.html

new marine industries and providing a solid foundation for marine new-quality productivity. In this context, analyzing the spatiotemporal evolution and the inherent patterns of the marine new-quality productivity is not only vital to meeting national strategic demands but also represents a theoretical breakthrough for addressing the structural contradictions of the marine economy.

From a global perspective, traditional marine economic research has primarily focused on economic benefits, such as input–output analyses in fisheries and shipping (Ji & Li, 2021). With the rise of sustainable development, the concept of 'blue economy', formally introduced at the Rio + 20 Conference in 2012, emphasizes marine environmental governance and the green transformation of marine industries. This highlights the coordinated development of ecological protection and economic growth (Liu et al., 2022; Ren et al., 2024a, b). The global digital wave has had a significant impact on marine resource allocation through digital technologies and data integration (Sun & Song, 2021). Scholars worldwide increasingly view marine technological innovation as key to marine economic growth. Liu et al. (2021) found a nonlinear relationship between marine technology innovation and the green total factor productivity of the marine economy. Zhang et al. (2025) examined the impact of the digital economy on marine fisheries' carbon emissions. Overall, research on the global marine economy is shifting from traditional single-dimensional analyses to a multi-faceted 'ecology–economy–technology' framework. These transformations represent an important aspect of marine new-quality productivity and are key drivers of high-quality marine economic development.

The concept of new-quality productivity has attracted much academic attention. The extensive economic growth model, driven by resource consumption, labor intensity, and capital input, no longer meets contemporary demands. This has created opportunities for fostering new-quality productivity (Wang et al., 2018). Several scholars regard new-quality productivity as an inherent evolution of traditional productivity. It emphasizes the principle of building before dismantling, giving due consideration to both new and old, with stability and advancement (Liu & He, 2024). As the marine economy enters a new stage of high-quality development, scholars are exploring how to harness new-quality productivity in the marine sector. Marine technological innovation is the core driving force. The extensive use of deep-sea exploration robots, intelligent aquaculture cages, and marine big data has broadened the approaches for deep-sea resource exploitation, transforming the traditional resource-dependent growth pattern (Huan et al., 2024; Liu et al., 2021). The modern marine industry system is marked by increasing sophistication, including transforming marine fishery and transportation toward high-end, intelligent development. Meanwhile, strategic industries such as marine new energy, marine biomedicine, and marine new materials are constantly emerging (Chen et al., 2025; Liu et al., 2023; Sepehri et al., 2024). Marine new-quality productivity also embodies the concept of green development, facilitating marine ecological restoration and protection using advanced technologies. Consequently, marine ecological governance has improved and promoted harmonious coexistence

between humans and the ocean (Lin, 2020; Ren et al., 2024a, b). Currently, there is no consensus on the definition of marine new-quality productivity, which warrants further research.

Marine new-quality productivity is an important part of new-quality productivity. It represents the innovative focus and future development trends of the marine economy. However, problems such as its measurement and evolution mechanisms still warrant in-depth studies. Currently, scholars primarily use comprehensive evaluation methods to assess the development of the marine economy, including subjective, objective, and combined weighting methods (Pan et al., 2024; Sun et al., 2023; Wu et al., 2024). Although the research methodology is fairly standard, there are obvious weaknesses in the choice of indicators. Existing studies have not sufficiently captured the characteristics of the transformation and upgrading of marine industries and the increase in marine productivity. Moreover, the main feature of marine new-quality productivity is a significant increase in marine total factor productivity, reflecting improvements in the quality and efficiency of economic development. Some scholars have calculated it using a DEA model. However, the choice of indicators is still limited to traditional factors such as input and output indicators, including labor, capital, environment, and resources. This traditional framework struggles to comprehensively reflect the overall state of marine economy development (Ding et al., 2020; Guo et al., 2022). It is clear that existing is still largely embedded in the evaluation system for the high-quality development of the marine economy (Sun et al., 2024b). Therefore, a comprehensive consideration of the multidimensional basis for the energy-level leap and for improving the quality and efficiency of marine productivity is needed to effectively assess marine new-quality productivity. Simultaneously, a set of indicators should be developed, with breakthroughs in marine science and technology as a key driver, innovative allocation of elements as a driver, and the industrial system as a vehicle, defining the characteristics of the two dimensions of the marine industry stock 'renovation' and 'transition' of the increment.

However, existing research still has limitations. First, at the level of conceptual development, most literature opts for replicating the framework of marine new-quality productivity. Such an approach does not fully reflect the complexity and specificity of the marine economy. Second, as regards quantitative analysis, recent research has emphasized the assessment of high-quality development of the marine economy. However, the complex circumstances of marine industrial transformation and the increase in marine productivity are not reflected in the indicator system. Third, the assessment perspective of the spatiotemporal evolution characteristics of marine new-quality productivity is rather simplistic and awaits further studies.

Accordingly, this study presents three main contributions. First, an integrated evaluation index framework is constructed based on the three-dimensional reconfiguration of 'New laborers, New labor materials, and New labor objects'. This framework enables the effective evaluation of marine new-quality productivity. Second, from the economic geography perspective, the spatiotemporal evolution patterns and future evolutionary tendencies of marine new-quality productivity should be disclosed. Third, by concentrating on the marine resource endowment and the foundation of marine economic development in China's coastal regions, regionally differentiated development paths in accordance with local conditions should be explored.

2 Methods and data

2.1 Study areas and data sources

This 2010–2023 study focused on 11 coastal province-level administrative divisions of China. Based on the division criteria set by the Ministry of Natural Resources of the People's Republic of China, the coastal area was segmented into three marine economic circles: The northern circle (Liaoning, Hebei, Tianjin, Shandong), the eastern circle (Jiangsu, Shanghai, Zhejiang), and the southern circle (Fujian, Guangdong, Guangxi, Hainan).

Data were obtained from the China Marine Economy Statistical Yearbook, China Environmental Statistical Yearbook, China Torch Statistical Yearbook, China Statistical Yearbook, Bulletin on the Status of China's Marine Ecological Environment, and other relevant official sources over successive years. Some missing data were filled using interpolation.

2.2 Construction of an evaluation index system

Marine economy largely depends on resources, high environmental sensitivity, and high technological content, making marine new-quality productivity distinct from traditional productive forces. Therefore, this study constructs a marine new-quality productivity indicator system from three dimensions: Laborers, labor materials, and labor objects, as shown in Table 1 (Chen et al., 2024; Sun & Song, 2021; Wang et al., 2023).

New laborers drive marine new-quality productivity (Han et al., 2024). Traditional marine economies depend on labor-intensive work, such as fishing and aquaculture. In contrast, developing marine new-quality productivity needs a high-caliber, specialized, and innovation-oriented workforce. Indicators are chosen based on quantitative measures of workforce scale and qualitative indicators for structural optimization.

Table 1 Index system of marine new-quality productivity

Constituent elements	Primary indicators	Secondary indicators	Tertiary indicators	Weights
New laborers	Quantity	Scale	Marine science and technology activity personnel with a master's degree or above	0.036
			Graduates with a master's degree or above in marine-related fields	0.029
	Quality	Structure	Proportion of senior professional titles in marine research institutions	0.005
			Per capita expenditure on marine education①	0.022
New labor objects	Expansion of marine resource acquisition	Utilization	Economic density of the coastline	0.060
			Cumulative installed capacity of offshore wind power	0.125
			Output of distant-water fishery	0.047
	Transformation of industrial systems	Upgrading	Area of national-level marine ranching demonstration zones	0.093
			Number of national-level marine industrial parks	0.019
			Rationalization index of the marine industry②	0.007
			Advanced index of the marine industry③	0.013
			Proportion of marine scientific research, education, management and the service industry	0.014
	Greening of industrial development	Greening	Volume of wastewater discharged directly into the sea from pollution sources	0.003
			Area of coastal and estuarine wetlands	0.027
			Construction situation of national marine nature reserves in China	0.113
			Investment in marine economic pollution control and treatment	0.030
New labor materials	Material means	Traditional infrastructure	Number of coastal observation stations	0.024
			Number of berths for production use at ports with a capacity of over 10,000 tons	0.012
			Number of first-class fishing ports	0.018
			Number of marine oil and gas production wells	0.076
		Digital infrastructure	Number of national-level science and technology business incubators along the coast	0.035
			Number of domain names in coastal areas	0.039
			Number of coastal Internet broadband access ports	0.021
	Immaterial means	Technological innovation	Number of marine research institutions	0.016
			Number of authorized invention patents in marine science and technology	0.041
			Internal expenditure on marine R&D funds	0.038
		Digital transformation	Digital inclusive finance index	0.013
			Penetration rate of broadband Internet	0.013
			Proportion of information transmission, software and information technology services in the total fixed asset investment of the whole society	0.013

① Per capita expenditure on marine education = local fiscal education expenditure × (regional GOP/regional GDP)/population size;

② Rationalization index of marine industry = marine tertiary industry output/(marine secondary industry output + the output value of marine primary industry);

③ Advanced index of marine industry = industrial height × industrial proportion (the proportion of the output value of the primary, secondary, and tertiary marine industries in the marine gross domestic product is multiplied by 1, 2, and 3, respectively, and then added together)

New labor objects, as the basis of productive activities, represent a qualitative shift beyond traditional marine labor. The expansion of labor objects occurs with the constructing of a modern marine industrial system as a significant carrier. It is a dynamic process from resource acquisition to value-added production and future-oriented industrial development. Evaluation metrics are organized into three dimensions: Marine resource expansion (stage one), industrial transformation (stage two), and greening of industrial development (stage three). This framework integrates sustainability, economic contributions, and environmental constraints (Wang et al., 2023). New labor materials represent multidimensional innovations in marine productivity. They include both tangible productive assets (material carriers) and intangible conduits (technological mediators) that transform value-creation processes in two synergistic ways. The tangible means used to change labor objects are a combination of modernized traditional infrastructure and new labor materials, such as the emerging digital infrastructure. In contrast, intangible means are mainly achieved through technological innovation and digital transformation (Liu et al., 2021; Wang et al., 2025).

2.3 Research methods

2.3.1 Entropy method

The entropy weighting method is based on information entropy theory. It determines the weights by computing the indicators' degree of discreteness and can effectively quantify the significance of each indicator in a complex system (Qin & Guo, 2025). The calculation methods are as follows.

First, the original data matrix $X = (x_{ij})_{m\times n}$ is standardized to eliminate dimensionality bias. The standardized matrix is $Z = (z_{ij})_{m\times n}$.

$$z_{ij} = \frac{x_{ij} - \min(x_j)}{\max(x_j) - \min(x_j)} (i = 1, 2, \cdots, m; j = 1, 2, \cdots, n) \tag{1}$$

$$z_{ij} = \frac{\max(x_j) - x_{ij}}{\max(x_j) - \min(x_j)} (i = 1, 2, \cdots, m; j = 1, 2, \cdots, n) \tag{2}$$

Second, calculate the j_{th} indicator weight p_{ij} for i_{th} solution.

$$p_{ij} = \frac{z_{ij}}{\sum_{i=1}^{m} z_{ij}} \tag{3}$$

Third, the information entropy e_j and the coefficient of variation d_j of indicator j are calculated.

$$e_j = -\frac{1}{\ln m} \sum_{i=1}^{m} p_{ij} \ln p_{ij} \tag{4}$$

$$d_j = 1 - e_j \tag{5}$$

Fourth, based on these outcomes, the weight w_j of indicator j is then computed.

$$w_j = \frac{d_j}{\sum_{j=1}^{n} d_j} \tag{6}$$

Fifth, the final total score is obtained.

$$MNQP_i = NL_i + NLO_i + NLM_i = \sum_{j=1}^{n} w_j z_{ij} \tag{7}$$

where $MNQP_i$ is the total score of marine new-quality productivity, which is the sum of the indices of new laborers (NL_i), new labor objects (NLO_i) and new labor materials (NLM_i).

2.3.2 Kernel density estimation

Kernel density estimation helps examine spatial non-equilibrium distributions. It defines the random variables' spatial distribution characteristics using continuous and smooth kernel density curves, which reflect distribution location, form, extensibility, and polarization phenomenon and show the dynamic evolution of spatial pattern distribution (Zhou et al., 2023).

$$f(x) = \frac{1}{nh} \sum_{i=1}^{n} K\left(\frac{X_j - x}{h}\right) \tag{8}$$

$$K(x) = \frac{1}{\sqrt{2\pi}} \exp\left(-\frac{x^2}{2}\right) \tag{9}$$

where $f(x)$ indicates the estimated value of the kernel density; $K(x)$ constitutes the kernel function; n represents the number of provinces within the region; X_i is the independently and identically distributed development level of the new marine quality productivity; x is the average value of the new marine quality productivity; h constitutes the bandwidth.

2.3.3 Dagum Gini coefficient

The Dagum Gini coefficient is a dimensionless measure of the relative disparity in development among individuals (Li et al., 2024a). To define the spatial heterogeneity of marine new-quality productivity in China's coastal regions, we use the Dagum Gini coefficient method to analyze the magnitude, sources, and contribution rates

of its spatial disequilibrium. According to this theory, the overall Gini coefficient G can be decomposed into intra- and inter-regional disparities (G_w and G_{nb}, respectively) and hypervariable density G_t based on the sources of subgroups.

$$G = \sum_{j=1}^{k}\sum_{h=1}^{k}\sum_{i=1}^{n_j}\sum_{r=1}^{n_h}\left|y_{ji} - y_{hr}\right| / 2n^2\mu \tag{10}$$

$$G_w = \sum_{j=1}^{k} G_{jj} p_j s_j \tag{11}$$

$$G_{nb} = \sum_{j=2}^{k}\sum_{h=1}^{j-1} G_{jh}(p_j s_h + p_h s_j) D_{jh} \tag{12}$$

$$G_t = \sum_{j=2}^{k}\sum_{h=1}^{j-1} G_{jh}(p_j s_h + p_h s_j)(1 - D_{jh}) \tag{13}$$

2.3.4 Markov chains

A Markov chain is a stochastic process with state-to-state transitions. The Markov transition probability matrix shows the internal dynamics and evolution trends of China's marine new-quality productivity.

$$p\{X(t) = j | X(t-1) = i, X(t-2) = i_{t-2}, ..., X(0) = i_0\} = p\{X(t) = j | X(t-1) = i\} \tag{14}$$

The state of the random variable X at a future time t is solely influenced by the previous period, implying that X has the first-order Markov property. The evolution of marine new-quality productivity distribution can be inferred from the state transition matrix p, which contains transition probabilities. Adding the spatial weight matrix yields a spatial Markov chain, combining the traditional Markov chain with spatial lag, to analyze how neighboring regions influence local transfers (Li et al., 2024b).

3 Results and discussions

3.1 Temporal evolutionary characteristics

Figure 2 shows the temporal changes in the average marine new-quality productivity and its components in China's coastal regions. The marine new-quality productivity in China is less developed, which indicates its nascent stage. However, a steady upward trend, also seen in the provinces, suggests rapid policy- and technology-driven innovation and progress. Table 2 shows increasing annual levels of marine new-quality productivity across coastal administrative divisions, suggesting progress. The provinces of Hebei and Hainan, along with the Guangxi Zhuang autonomous region, lag behind other coastal areas due to their less robust foundation for a marine economy (Di et al., 2022).

The development levels of each element consistently grew (Fig. 2). The new labor index grows slowly, indicating a structural imbalance in marine labor. Cultivating high-caliber, innovative talent to boost marine productivity is complex and long-term. The lack of high-end talent hampers the growth of the new labor index, whereas traditional marine workers, such as fishermen, struggle with skill transformation to meet modern demands. Advancements in marine technological innovation and digital transformation have rapidly increased the index of new labor materials, showing significant improvements

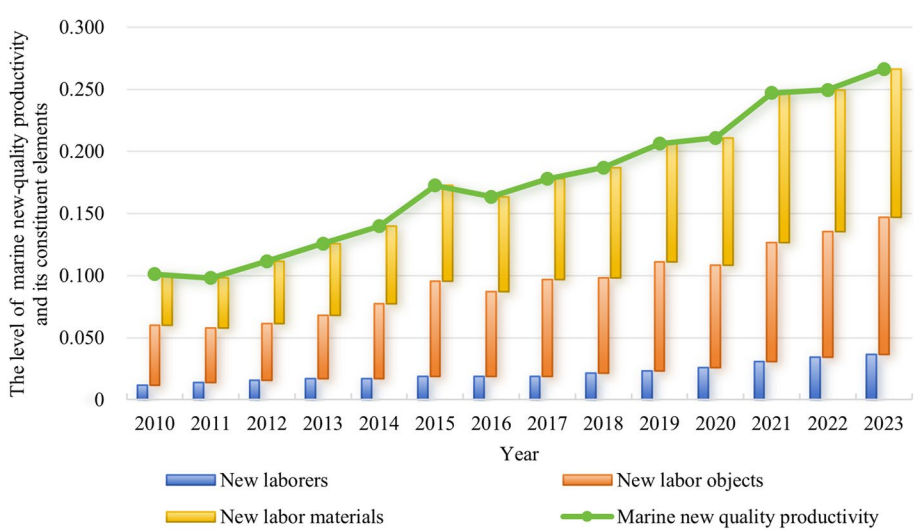

Fig. 2 Temporal evolutionary characteristics of China's marine new-quality productivity and the levels of its constituent elements

Table 2 Temporal evolutionary characteristics of marine new-quality productivity in China's coastal areas

	2010	2011	2012	2013	2014	2015	2016	2017	2018	2019	2020	2021	2022	2023
Tianjin	0.118	0.128	0.142	0.159	0.174	0.192	0.179	0.188	0.196	0.209	0.197	0.223	0.151	0.158
Hebei	0.047	0.054	0.063	0.072	0.083	0.093	0.105	0.111	0.138	0.145	0.135	0.159	0.161	0.164
Liaoning	0.102	0.102	0.121	0.144	0.154	0.182	0.160	0.160	0.163	0.175	0.174	0.202	0.191	0.202
Shanghai	0.135	0.146	0.161	0.173	0.190	0.208	0.211	0.214	0.213	0.232	0.243	0.279	0.287	0.300
Jiangsu	0.118	0.087	0.101	0.124	0.132	0.162	0.169	0.225	0.213	0.234	0.244	0.317	0.323	0.337
Zhejiang	0.083	0.093	0.123	0.128	0.149	0.183	0.179	0.185	0.196	0.200	0.217	0.237	0.253	0.279
Fujian	0.084	0.085	0.097	0.114	0.125	0.260	0.157	0.187	0.195	0.198	0.188	0.220	0.227	0.224
Shandong	0.144	0.152	0.172	0.198	0.230	0.245	0.257	0.257	0.278	0.293	0.300	0.340	0.385	0.429
Guangdong	0.127	0.132	0.157	0.173	0.185	0.246	0.256	0.281	0.308	0.385	0.401	0.491	0.517	0.559
Guangxi	0.036	0.041	0.031	0.033	0.048	0.050	0.051	0.065	0.064	0.072	0.084	0.095	0.093	0.101
Hainan	0.120	0.058	0.061	0.065	0.067	0.076	0.077	0.084	0.093	0.127	0.135	0.154	0.155	0.177

in 'hard power' elements such as infrastructure and technical equipment. The index of new labor objects also increases by optimizing industrial parks and exploring new energy sources. Overall, the contribution of new labor objects and materials to productive forces is rising, whereas that of the labor force index declines annually. This suggests that technology is increasingly replacing human resources as the main driver of maritime economy development.

Accordingly, Fig. 3 shows the analysis of the evolution of China's marine new-quality productivity kernel density. The distribution curve's center has continuously shifted to the right, which validates our earlier findings of steadily improving China's marine new-quality productivity. The distribution patterns show that the main peak's height shows a fluctuating but declining trend of 'N'. The distribution bandwidth shows a typically narrowing pattern of 'narrowing–expansion–narrowing', which suggests a gradually decreasing dispersion in the development of marine new-quality productivity. The distribution curve shows a right-tailed phenomenon that gradually diminishes, indicating a tendency toward contraction. Some province-level administrative divisions have much higher marine new-quality productivity levels, while lower-level administrative divisions are approaching higher levels after adjustments. From the perspective of the peak quantity evolution, the curve shifts from unimodal to bimodal with 'one main and one side', and then gradually returns to unimodal over time. This suggests that multi-level differentiation in marine new-quality productivity initially intensifies and then lessens, thereby reducing internal differences and disparities in regional development.

3.2 Spatial evolutionary characteristics

Marine new-quality productivity has charted a new direction for promoting coordinated development across China's coastal areas. Figure 4 shows the levels of marine new-quality productivity in the three major marine economic circles (i.e., northern, eastern, and southern) from 2010 to 2023. Overall, all three circles show a clear upward trend, with the eastern marine economic circle maintaining a leading position throughout the period (Zhao et al., 2023). The eastern region has long invested heavily in marine science and technology innovation, high-end manufacturing, and emerging marine industries. As a leader in China's marine economic development, it has attracted considerable talent and capital. Since the 13th Five-Year Plan, the state has increasingly prioritized the marine economy and implemented a series of supportive policies. Leveraging its first-mover advantage, the eastern region has further strengthened investment and innovation in the marine field, thereby increasing its lead over other regions. The northern regional economic circle lags behind the east, with a trend largely consistent with national variations in marine new-quality productivity. This indicates that its development is influenced by national maritime economic conditions while maintaining a relatively stable development of its own. The northern region has actively advanced maritime resource development, industrial upgrading, and marine ecological protection. Through

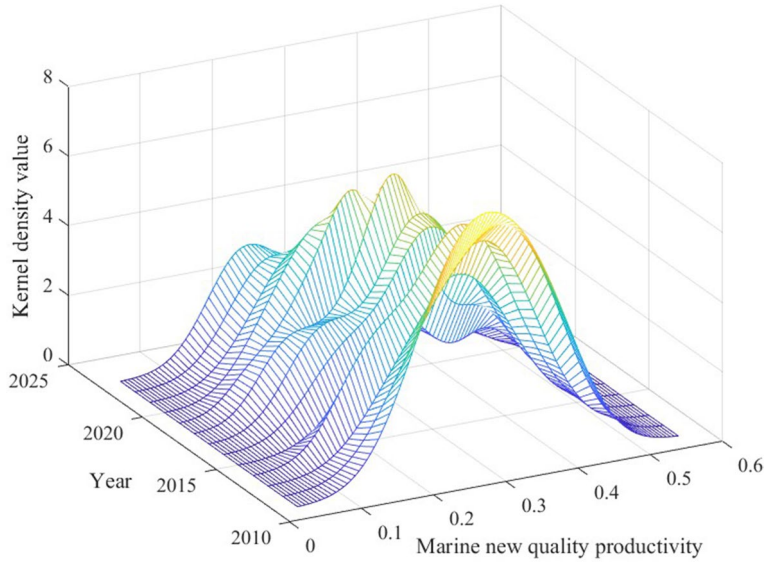

Fig. 3 Kernel density estimation and dynamic evolution of marine new-quality productivity in China

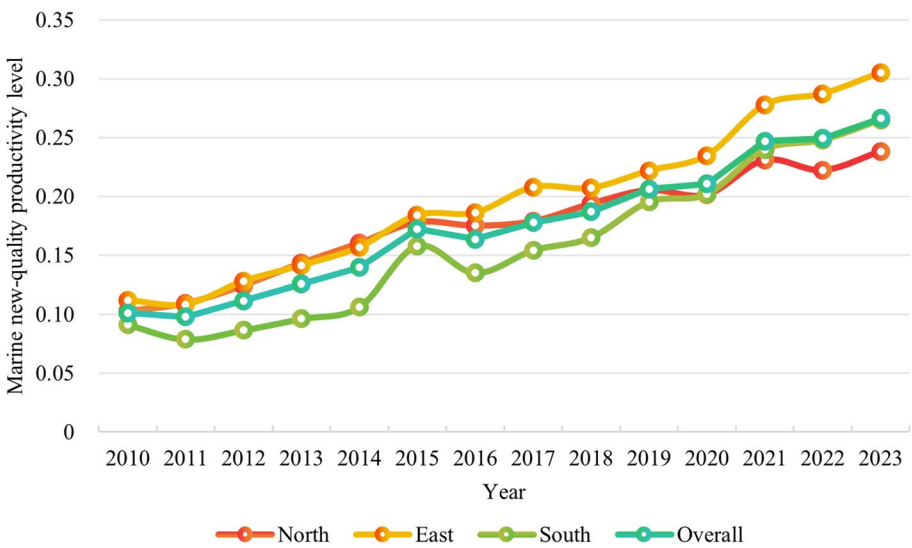

Fig. 4 Temporal evolution trend of the marine new-quality productivity in China's three major marine economic circles

industrial restructuring, enhanced regional cooperation, and scientific marine innovations, it has achieved steady growth in new-quality productivity. From 2010 to 2020, the southern marine economic circle consistently ranked lowest, likely due to challenges such as regional development imbalances in the course of marine economic development. Among them, Guangdong province has consistently maintained a leading position, benefiting from a favorable geographical location and distinct industrial advantages. However, Guangxi and Hainan are comparatively disadvantaged, with a weaker foundation for developing the marine economy. Since entering the 14th Five-Year Plan period, the southern marine economic circle has increasingly shown off its late-comer advantages, gradually overtaking the northern marine economic circle.

We further used radar diagrams to show the movement and variation of the constituent elements of the marine new-quality productivity across regions (Fig. 5). These patterns reflect the comparative advantages of each marine economic circle in developing marine new-quality productivity. The new labor index shows low levels and minimal growth in the three major marine economic circles. This indicates that the labor

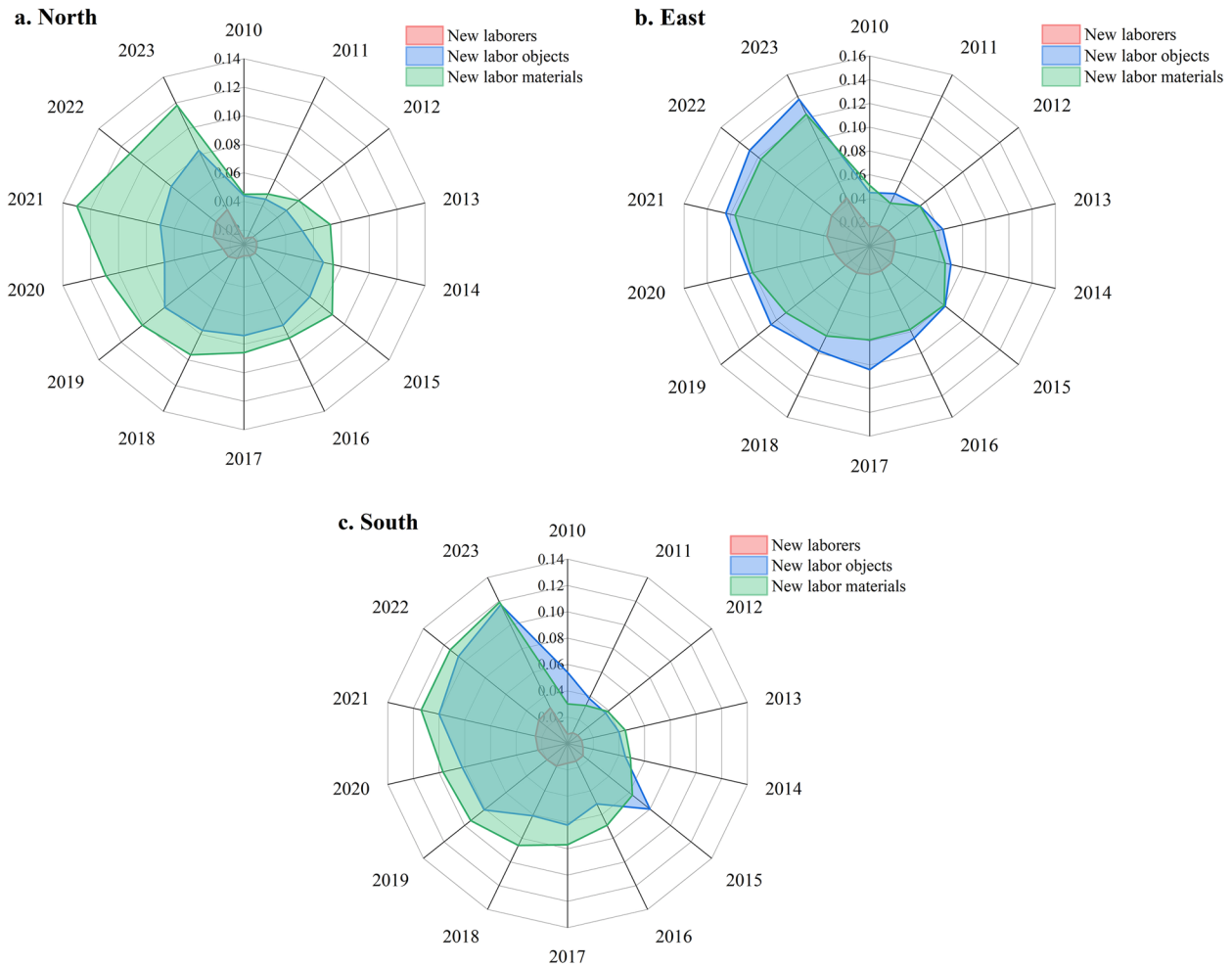

Fig. 5 Changes in the constituent elements of the marine new-quality productivity in China's three major marine economic circles

force quality and structure remain long-term constraints on the development of new-quality productivity in the marine sector. This finding aligns with the 'structural contradiction of employment' observed in China's coastal regions. The availability of high-skilled talent needed for strategic marine industries remains limited, and the shift of traditional labor is sluggish. In terms of the new labor materials index, all regions showed notable improvement over the study period, with values more than doubling. In particular, the southern marine economic circle, which was initially low, showed a threefold increase by 2023. This indicates significant progress in marine economic infrastructure development and technological innovation. In terms of the new labor objects index, the eastern marine economic circle showed the most remarkable growth, consistent with the 'core–periphery' theory. Leveraging its advantageous location, sophisticated infrastructure, and strong scientific research capacity, the eastern region has become China's core zone for marine technological innovation. This differentiated development pattern highlights the resource advantages of each economic circle while revealing the deficiencies in regional collaborative innovation. In the future, enhancing cross-regional factor flows and policy convergence will be crucial for realizing the potential of marine new-quality productivity.

3.3 Unbalanced regional pattern and causes

Grouped by the three major marine economic circles, this study further examines regional spatial disparities in marine new-quality productivity. Figure 6 shows the decomposition results of the Dagum Gini coefficient of China's marine new-quality productivity from 2010 to 2023.

The overall Gini coefficient remained at approximately 0.2 from 2010 to 2023, showing a phased pattern of 'rise–decline–rise'. This indicates that spatial disparities in China's marine new-quality productivity have fluctuated,

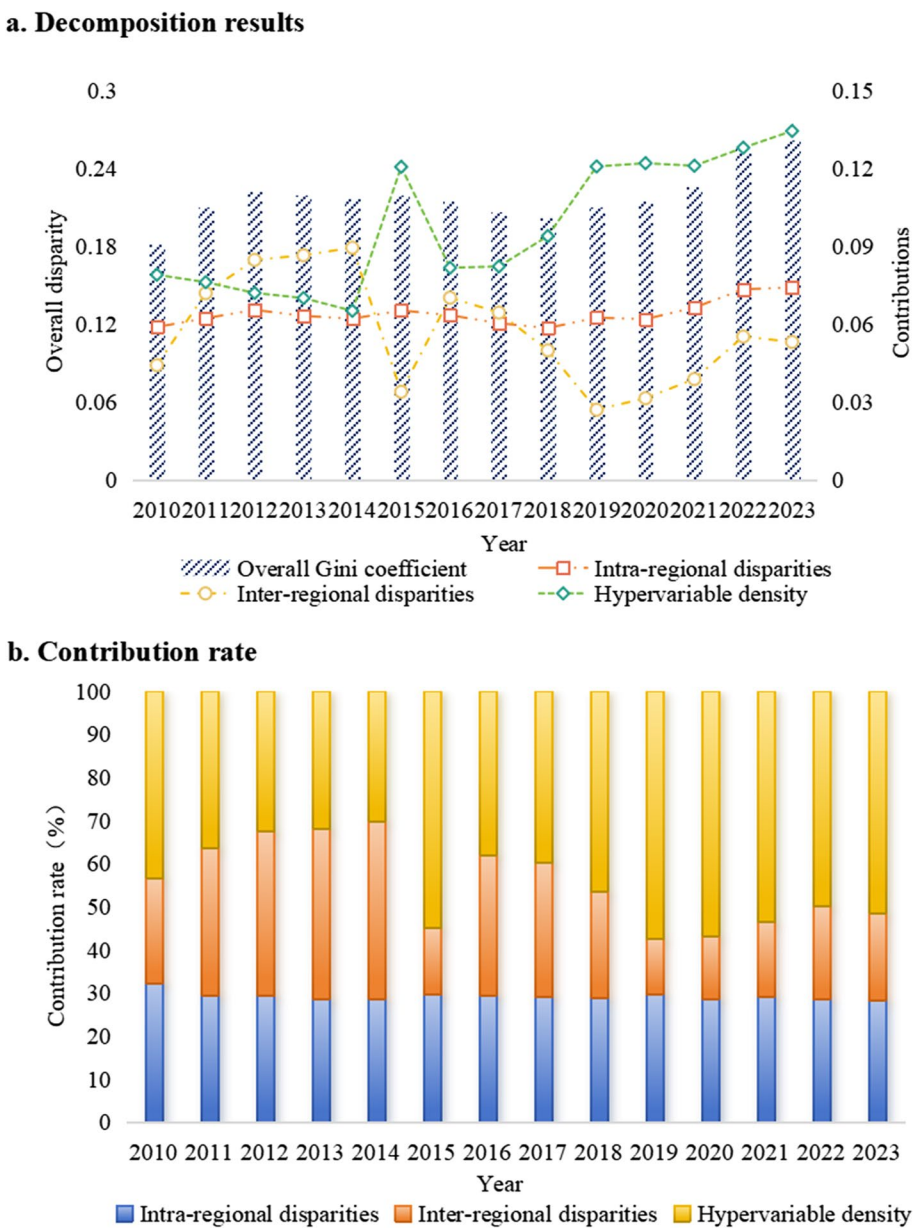

Fig. 6 Decomposition results of the regional differences of marine new-quality productivity in China

showing a trend of 'expansion–contraction–expansion'. Regional imbalances in development remain prominent. In terms of contributions to regional differences, differences in the development of marine new-quality productivity mainly originate from hypervariable density. This suggests that the growing complexity of cross-regional factor flows widens regional disparities.

The southern region exhibits the largest intra-regional disparity, followed by the north, while the east records the smallest (Sun et al., 2024a). The substantial resource endowment and economic foundation held by Guangdong and Fujian provinces attract a substantial labor force to converge locally, thereby using their factor advantages to further widen the marine economic gap with other provinces. Such imbalanced development may have gradually widened marine economic disparities in this area. Intra-regional disparity in both the northern and eastern marine economic circles remains relatively low. Although there have been occasional fluctuations, there is an overall declining trend. This suggests that the differences in marine new-quality productivity in these two regions are narrowing and improving steadily.

The northern and southern regions show the largest regional disparities, followed by the eastern and southern regions. Both are significantly greater than the disparity in the northeastern region. The north–south gap arises from a structural contradiction between traditional path dependence and the emerging dynamic forces. The 'North-heavy and South-light' marine industrial pattern hinders cross-regional factor flows, with labor and capital migrating only from north to south, with no reverse technology feedback mechanism. This results in a 'pump effect'. The eastern marine economic circle has benefited from the Yangtze River Delta, with coordinated development across marine engineering equipment, marine biomedicine, and digital shipping sectors. However, the southern region suffers from weak internal linkages and persistent unbalanced development.

This suggests 'club convergence', where similar province-level administrative divisions form a distinct 'club' within which convergence occurs, but diverge from other clubs. Specifically, province-level administrative divisions such as Shanghai, Jiangsu, and Zhejiang share comparable initial conditions, resource endowments, and policy orientations and form a high-level equilibrium 'club'. In contrast, province-level administrative divisions with weaker foundations, such as Guangxi and Hainan, are entrenched in a low-level development trajectory, resulting in a 'low-level lock-in' effect. This suggests that China's marine new-quality productivity is influenced by the 'Matthew Effect', where high-end factors, such as talents, capital, and technology, continue to agglomerate in the eastern region. However, the spillover and trickle-down effects on parts of the southern and northern regions remain too weak to overcome geographical distances and institutional barriers.

3.4 Forecasting of evolutionary trends

This study uses the traditional and spatial Markov chains to examine the transfer direction and state transition probabilities of China's marine new-quality productivity across coastal province-level administrative divisions. Based on the quartiles of the development levels of marine new-quality productivity, these administrative divisions are classified into four categories: Low (I), medium–low (II), medium–high (III), and high (IV).

Table 3 shows that the diagonal probabilities of the transfer matrix are significantly higher than those on the other non-diagonal lines. This indicates low internal mobility of the distribution of China's marine new-quality productivity levels across coastal areas, but remains highly stable. However, it is difficult to achieve cross-state transitions. When the coastal province-level administrative divisions' marine new-quality productivity is at level IV, there is a higher probability of remaining in the original state in the next stage. Meanwhile, for those in adjacent administrative divisions, cross-state transfer to higher- or lower-level administrative divisions s is less likely. For types II and III, there is a higher probability of upward transfer than that of downward transfer, suggesting that the marine new-quality productivity is often higher in regional dynamic evolution.

Table 3 Transition probability matrix of marine new-quality productivity in China

Type	I	II	III	IV	Number
I	0.84	0.16	0	0	38
II	0.05	0.62	0.30	0.03	37
III	0	0.03	0.75	0.22	36
IV	0	0.06	0.03	0.91	32

Table 4 shows that by adding the spatial lag condition, the geographical spatial context significantly influences the transfer process of the marine new-quality productivity levels. First, across all spatial lag types, the diagonal probabilities are consistently greater than those in the same row. This confirms the stability of marine new-quality productivity level transfers. Second, 'low-level lock-in' and 'high-level monopolization' effects are evident. High-level province-level administrative divisions exhibit a strong tendency to maintain their current state, with a low probability of a downward transition. This indicates that talents, venture capital, and key technologies tend to circulate within these cores rather than transfer to less developed regions, showing strong path dependence and spatial stickiness. Meanwhile, for low-level province-level administrative divisions, the probabilities of remaining at their current state are 0.89, 0.77, 0.80, and 1.00 when their neighboring areas are of low, medium–low, medium–high, and high levels, respectively. Third, the types of new-quality productivity levels in neighboring marine areas ease the transfer of provincial-level types. High-level neighbors improve surrounding provinces through a radiating effect. This highlights the geographical limitations of 'knowledge spillover'. Specifically, innovation is easily diffused among geographically and culturally proximate regions.

4 Conclusion

The concept of marine new-quality productivity aligns with the development needs of the marine economy. Using data from 11 coastal provinces in China from 2010 to 2023, this study assesses the development level of marine new-quality productivity and examines its spatio-temporal evolution patterns.

The study findings are as follows: First, during the study period, marine new-quality productivity in China's

Table 4 Spatial Markov transfer probability matrix of marine new-quality productivity in China

Type	t/t + 1	I	II	III	IV	Number
I	I	0.89	0.11	0	0	18
	II	0.07	0.71	0.21	0	14
	III	0	0	0.83	0.17	6
	IV	0	0	0	0	0
II	I	0.77	0.23	0	0	13
	II	0.08	0.50	0.33	0.08	12
	III	0	0.09	0.73	0.18	11
	IV	0	0.20	0	0.80	5
III	I	0.80	0.20	0	0	5
	II	0	0.71	0.29	0	7
	III	0	0	0.85	0.15	13
	IV	0	0	0	1.00	9
IV	I	1.00	0	0	0	2
	II	0	0.50	0.50	0	4
	III	0	0	0.50	0.50	6
	IV	0	0.06	0.06	0.89	18

coastal regions continued to grow steadily, with all components showing positive growth. The new labor materials index grew most rapidly, whereas that of new laborers remained stable. Second, the spatial pattern of 'eastern marine economic circle > northern marine economic circle > southern marine economic circle' persisted, highlighting significant regional development imbalances. Third, intra-regional disparity is highest within the southern marine economic circle, while inter-regional disparities between the northern–southern and eastern–southern marine economic circles are even higher. These disparities primarily arise from hypervariable density, indicating that complex cross-regional factor flows increasingly shape regional disparities. Fourth, the development of marine new-quality productivity is gradual, with its distribution showing strong stability and a tendency toward upward transfer.

The study findings offer new empirical evidence for the coordinated development of China's coastal areas and improved marine new-quality productivity. The policy implications are as follows: First, marine scientific and technological innovation should be boosted to optimize the combination of production factors. This includes nurturing marine scientific and technological innovation talent, improving the allocation of national marine strategic scientific and technological resources, fostering a dynamic scientific research ecosystem, and enhancing key factors such as marine laborers and labor inputs. Second, the spatial layout of marine regions should be continuously optimized by tailoring strategies to local conditions and exploring differentiated practices. Based on the actual regional development, marine economic development plans should align with the future trends of marine development and reflect local marine characteristics. Third, the radiating effects of advantaged regions should be actively leveraged by using the scientific, technological, talent, and resource strengthens of high-level province-level administrative divisions to strengthen linkages with neighboring administrative divisions, and thereby promote overall marine productivity. This approach should first narrow intra-regional disparities and then drive broader regional enhancement.

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Authors' contributions
All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Xin Zhao, Ping Gao and Zaixun Jia. The first draft of the manuscript was written by Ping Gao and Zaixun Jia and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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Data availability
The data that support the findings of this study are available from the corresponding author upon reasonable request.

Declarations

Ethics approval and consent to participate
Not applicable. This article does not contain any studies with human participants.

Competing interests

The authors declare that they have no competing interests. Xin Zhao is the Editor-in-Chief of this journal.

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